

List of Publications

01

PUBLISHED AND ACCEPTED RESEARCH

— Monographs

1. E. Emmrich and **A. Kaltenbach**. *Funktionen mit Werten in einem Banach-Raum: Funktionenräume für zeitabhängige Probleme*, **Springer-Lehrbuch**, 2026.
DOI 10.1007/978-3-662-72999-1
2. **A. Kaltenbach**. *Pseudo-monotone operator theory for unsteady problems with variable exponents*, **Lecture Notes in Mathematics**, 2023.
DOI 10.1007/978-3-031-29670-3

— PhD Thesis

3. **A. Kaltenbach**. *Theory of pseudo-monotone operators for unsteady problems in variable exponent spaces*, **Freidok University of Freiburg**, 2021.
DOI 10.6094/UNIFR/222538

— Journal Articles

4. **A. Kaltenbach**, H. Antil, and K. L. A. Kirk. *Convection Effects and Optimal Insulation: Modelling and Analysis*, **ESAIM: Control, Optimisation and Calculus of Variations**, published online, 2026.
DOI 10.1051/cocv/2026064
5. L. C. Berselli and **A. Kaltenbach**. *Pulsatile Flows for Simplified Smart Fluids with Variable Exponent Power-Law: Analysis and Numerics*, **Mathematical Models and Methods in Applied Sciences**, published online, 2026.
DOI 10.1142/S0218202527500023
6. **A. Kaltenbach** and J. Wichmann. *A Priori Error Analysis for the p -Stokes Equations with Slip Boundary Conditions: A Discrete Leray Projection Framework*, **IMA Journal of Numerical Analysis**, accepted, 2026.
DOI 10.48550/arXiv.2507.15016
7. H. Antil, **A. Kaltenbach**, and K. Kirk. *Duality-based algorithm and numerical analysis for optimal insulation problems on non-smooth domains*, **SIAM Journal on Control and Optimization**, 64(3), 1235–1268, 2026.
DOI 10.1137/25M1758647
8. J. Blechta, P. A. Gazca-Orozco, **A. Kaltenbach**, and M. Růžička. *Quasi-optimal discontinuous Galerkin discretisations of the p -Dirichlet problem*, **Numerische Mathematik**, 158(2), 587–626, 2026.
DOI 10.1007/s00211-025-01507-1
9. **A. Kaltenbach** and M. Zeinhofer. *The Deep Ritz Method for parametric p -Dirichlet problems*, **Advances in Continuous and Discrete Models**, 2026(1), Article 8, 2026.
DOI 10.1186/s13662-025-04043-2
10. H. Antil, **A. Kaltenbach**, and K. Kirk. *Modeling and Analysis of an Optimal Insulation Problem on Nonsmooth Domains*, **SIAM Journal on Mathematical Analysis**, 58(1), 450–470, 2026.
DOI 10.1137/25M1741947
11. S. Bartels and **A. Kaltenbach**. *Explicit a posteriori error representation for variational problems and application to TV-minimization*, **Foundations of Computational Mathematics**, 26(1), 131–177, 2026.
DOI 10.1007/s10208-024-09676-5
12. H. Antil, S. Bartels, **A. Kaltenbach**, and R. Khandelwal. *Variational problems with gradient constraints: A priori and a posteriori error identities*, **Mathematics of Computation**, published online, 2025.
DOI 10.1090/mcom/4146
13. S. Bartels, T. Gudi, and **A. Kaltenbach**. *A priori and a posteriori error identities for the scalar Signorini problem*, **SIAM Journal on Numerical Analysis**, 63(5), 2155–2186, 2025.
DOI 10.1137/24M1677691
14. S. Bartels and **A. Kaltenbach**. *Error analysis for a Crouzeix-Raviart approximation of the obstacle problem*, **Journal of Numerical Mathematics**, 34(3), 271–305, 2026.
DOI 10.1515/jnma-2025-0036

15. L. C. Berselli and **A. Kaltenbach**. *Convergence analysis of a fully-discrete finite element approximation of the unsteady $p(\cdot, \cdot)$ -Navier-Stokes equations*, **Numerische Mathematik**, 157(2), 573–627, 2025.
DOI 10.1007/s00211-025-01450-1
16. L. C. Berselli, **A. Kaltenbach**, and M. Růžička. *Energy conservation for weak solutions of incompressible Newtonian fluid equations in Hölder spaces with Dirichlet boundary conditions in the half-space*, **Mathematische Annalen**, 391(4), 5911–5940, 2025.
DOI 10.1007/s00208-024-03065-7
17. L. C. Berselli and **A. Kaltenbach**. *Error analysis for a finite element approximation of the steady $p(\cdot)$ -Navier-Stokes equations*, **IMA Journal of Numerical Analysis**, 45(5), 3026–3076, 2025.
DOI 10.1093/imanum/drae082
18. A. Kh. Balci and **A. Kaltenbach**. *Error analysis for a Crouzeix-Raviart approximation of the variable exponent Dirichlet problem*, **IMA Journal of Numerical Analysis**, 45(2), 1102–1142, 2025.
DOI 10.1093/imanum/drae025
19. P. A. Gazca-Orozco and **A. Kaltenbach**. *On the stability and convergence of Discontinuous Galerkin schemes for incompressible flows*, **IMA Journal of Numerical Analysis**, 45(1), 243–282, 2025.
DOI 10.1093/imanum/drae004
20. J. Jeřberger and **A. Kaltenbach**. *Finite element discretization of the steady, generalized Navier-Stokes equations with inhomogeneous Dirichlet boundary conditions*, **SIAM Journal on Numerical Analysis**, 62(4), 1660–1686, 2024.
DOI 10.1137/23M1607398
21. **A. Kaltenbach** and M. Růžička. *Note on quasi-optimal error estimates for the pressure for shear-thickening fluids*, **ESAIM: Mathematical Modelling and Numerical Analysis**, 58(5), 1907–1933, 2024.
DOI 10.1051/m2an/2024051
22. **A. Kaltenbach**. *Error analysis for a Crouzeix-Raviart approximation of the p -Dirichlet problem*, **Journal of Numerical Mathematics**, 32(2), 111–138, 2024.
DOI 10.1515/jnma-2022-0106
23. **A. Kaltenbach** and M. Růžička. *A Local Discontinuous Galerkin approximation for the p -Navier-Stokes system, Part III: Convergence rates for the pressure*, **SIAM Journal on Numerical Analysis**, 61(4), 1763–1782, 2023.
DOI 10.1137/22M1541472
24. **A. Kaltenbach** and M. Růžička. *A Local Discontinuous Galerkin approximation for the p -Navier-Stokes system, Part II: Convergence rates for the velocity*, **SIAM Journal on Numerical Analysis**, 61(4), 1641–1663, 2023.
DOI 10.1137/22M1514751
25. **A. Kaltenbach** and M. Růžička. *A Local Discontinuous Galerkin approximation for the p -Navier-Stokes system, Part I: Convergence analysis*, **SIAM Journal on Numerical Analysis**, 61(4), 1613–1640, 2023.
DOI 10.1137/22M151474X
26. **A. Kaltenbach** and M. Růžička. *Convergence analysis of a Local Discontinuous Galerkin approximation for nonlinear systems with balanced Orlicz-structure*, **ESAIM: Mathematical Modelling and Numerical Analysis**, 57(3), 1381–1411, 2023.
DOI 10.1051/m2an/2023028
27. L. C. Berselli, **A. Kaltenbach**, R. Lewandowski, and M. Růžička. *On the existence of weak solutions for a family of unsteady rotational Smagorinsky models*, **Pure and Applied Functional Analysis**, 8(1), 83–102, 2023.
28. **A. Kaltenbach** and M. Růžička. *Existence of steady solutions for a model for micropolar electrorheological fluid flows with not globally log-Hölder continuous shear exponent*, **Journal of Mathematical Fluid Mechanics**, 25(2), Article 40, 2023.
DOI 10.1007/s00021-023-00782-y
29. **A. Kaltenbach** and M. Růžička. *Analysis of a fully-discrete, non-conforming approximation of evolution equations and applications*, **Mathematical Models and Methods in Applied Sciences**, 33(6), 1147–1192, 2023.
DOI 10.1142/S0218202523500197
30. S. Bartels and **A. Kaltenbach**. *Explicit and efficient error estimation for convex minimization problems*, **Mathematics of Computation**, 92(343), 2247–2279, 2023.
DOI 10.1090/mcom/3821
31. **A. Kaltenbach** and M. Růžička. *Existence of steady solutions for a general model for micropolar electrorheological fluid flows*, **SIAM Journal on Mathematical Analysis**, 55(3), 2238–2260, 2023.
DOI 10.1137/22M1500599

32. S. Bartels and **A. Kaltenbach**. *Error estimates for total-variation regularized minimization problems with singular dual solutions*, **Numerische Mathematik**, 152(4), 881–906, 2022.
DOI 10.1007/s00211-022-01324-w
33. **A. Kaltenbach**. *Note on the existence theory for non-induced evolution equations*, **Mathematische Nachrichten**, 295(6), 1186–1210, 2022.
DOI 10.1002/mana.201900555
34. L. C. Berselli, **A. Kaltenbach**, and M. Růžička. *Analysis of fully discrete, quasi non-conforming approximation of evolution equations and applications*, **Mathematical Models and Methods in Applied Sciences**, 31(11), 2297–2343, 2021.
DOI 10.1142/S0218202521500494
35. **A. Kaltenbach** and M. Růžička. *Variable exponent Bochner-Lebesgue spaces with symmetric gradient structure*, **Journal of Mathematical Analysis and Applications**, 503(2), Article 125355, 2021.
DOI 10.1016/j.jmaa.2021.125355
36. **A. Kaltenbach** and M. Růžička. *Note on the existence theory for pseudo-monotone evolution problems*, **Journal of Evolution Equations**, 21(1), 247–276, 2021.
DOI 10.1007/s00028-020-00577-y

— Book Chapter

37. S. Bartels and **A. Kaltenbach**. *Exact a posteriori error control for variational problems via convex duality and explicit flux reconstruction*, **Advances in Applied Mechanics**, 58, 295–375, 2024.
DOI 10.1016/bs.aams.2024.04.001

— Conference Proceedings

38. **A. Kaltenbach** and M. Růžička. *Conditional quasi-optimal error estimate for a finite element discretization of the p -Navier-Stokes equations: the case $p > 2$* . In J.-i. Segata, H. Takeda, M. Mizuno, and M. Yamamoto (eds.), **Critical Phenomena in Nonlinear Partial Differential Equations, Harmonic Analysis, and Functional Inequalities, Vol. 1**, Tohoku Series in Mathematical Sciences, vol. 2, Springer Singapore, forthcoming, 2026.
DOI 10.48550/arXiv.2411.00043

02

PREPRINTS

Submitted manuscripts; not yet accepted for publication.

39. H. Antil, **A. Kaltenbach**, and S. Shah. *QH-GEM: Quantum-Hydrodynamic Generative Modeling*, submitted, 2026.
DOI 10.48550/arXiv.2608.27216
40. H. Antil, **A. Kaltenbach**, and K. Kirk. *Duality-Based A Posteriori Error Identities for Subgradient Flows Based on the Brézis-Ekeland-Nayroles Principle*, submitted, 2026.
DOI 10.48550/arXiv.2608.17845
41. S. Bartels and **A. Kaltenbach**. *A prox-Based Semi-Smooth Newton Method for Convex Variational Problems*, submitted, 2026.
DOI 10.48550/arXiv.2606.25948
42. H. Antil, **A. Kaltenbach**, and K. Kirk. *A Finite Element Approximation of an Optimal Insulation Problem with Convective Heat Transfer*, submitted, 2026.
DOI 10.48550/arXiv.2606.14991
43. S. Bartels and **A. Kaltenbach**. *A prox-Based Semi-Smooth Newton Method for TV-Minimization*, submitted, 2026.
DOI 10.48550/arXiv.2605.22728
44. P. A. Gazca-Orozco and **A. Kaltenbach**. *A priori and a posteriori error identities for vectorial problems via convex duality*, submitted, 2026.
DOI 10.48550/arXiv.2602.04368
45. L. C. Berselli, **A. Kaltenbach**, and S. Ko. *Error analysis for a fully-discrete finite element approximation of the unsteady $p(\cdot, \cdot)$ -Stokes equations*, submitted, 2025.
DOI 10.48550/arXiv.2501.00849
46. J. Jeßberger and **A. Kaltenbach**. *Finite element discretization of the steady, generalized Navier-Stokes equations with small shear exponent*, submitted, 2024.
DOI 10.48550/arXiv.2408.15731

27 September 2026.